

11 Eigenvalue Decomposition



11.1 Eigenvalue Decomposition

An eigenvalue is a matrix's "own value"; an eigenvector is its "own direction" — solving eigenvalue decomposition by hand

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master in This Section

1

An eigenvector is a nonzero vector whose direction is unchanged by a matrix transform

2

An eigenvalue is the factor by which the eigenvector is scaled

3

Solve eigenvalue decomposition by hand using the characteristic polynomial

4

All eigenvectors for one eigenvalue form a line through the origin

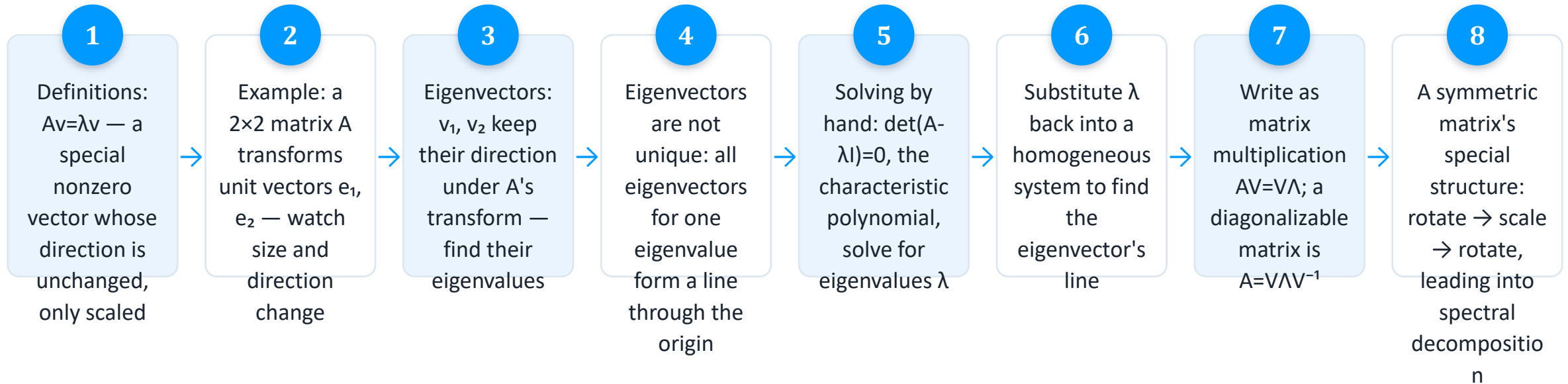
5

A matrix is diagonalizable exactly when its eigenvectors are linearly independent

6

The eigendecomposition of a real symmetric matrix is called spectral decomposition

From $Av=\lambda v$ to Solving by Hand, to Spectral Decomposition



Seven Lenses for Every Topic in This Chapter

① Geometric view: an eigenvector keeps its direction under a matrix transform; its eigenvalue is the scale factor

② Algebraic view: characteristic polynomial $\det(\mathbf{A}-\lambda\mathbf{I})=0$, diagonalization $\mathbf{A}=\mathbf{V}\mathbf{\Lambda}\mathbf{V}^{-1}$

③ Symmetric matrices: spectral decomposition $\mathbf{A}=\mathbf{V}\mathbf{\Lambda}\mathbf{V}^T$, geometrically "rotate $\rightarrow$ scale $\rightarrow$ rotate"

④ Asymmetric matrices: the decomposition is no longer orthogonal; eigenvectors need not be orthogonal

⑤ Applications: eigendecomposing the Gram matrix or covariance matrix underlies PCA

⑥ Coding view: `numpy.linalg.eig` computes eigenvalues and eigenvectors, and verifies the decomposition

⑦ Generalization path: from eigenvalue decomposition onward to singular value decomposition (SVD)

Eigendecomposition finds a matrix A 's eigenvalue λ and eigenvector v , satisfying

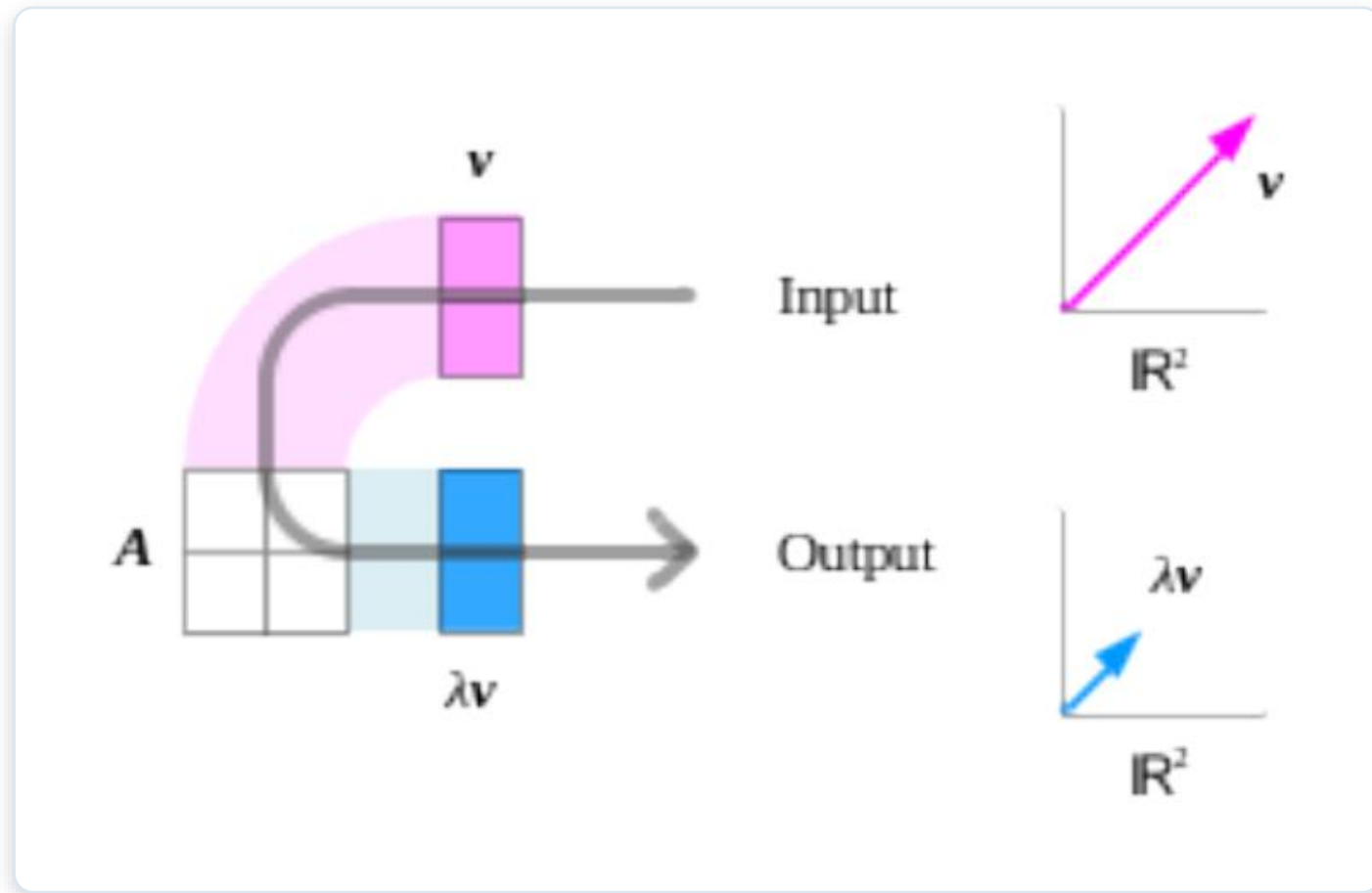
$$Av = \lambda v$$

- "Eigen" comes from German, meaning "own" or "inherent"
- A acting on v gives λv ; under A , v keeps its direction (or flips exactly to the opposite)

$$Av = \lambda v$$

Figure 1 · $Av = \lambda v$, Matrix A Is 2×2

- λ sets the scaling size; its sign decides whether the output stays aligned with v or flips



AN EXAMPLE

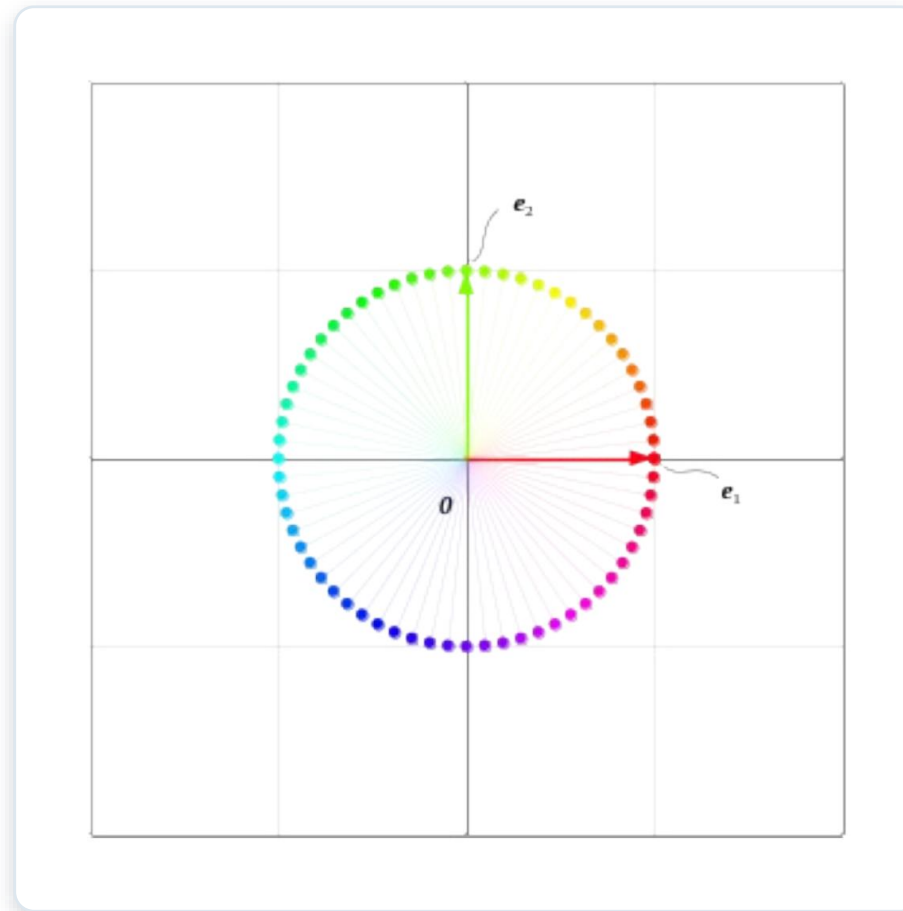
Given a 2×2 matrix A, apply it to transform the unit vectors shown

$$A = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

- Eigenvectors are assumed nonzero, since $A\mathbf{0}=\mathbf{0}$ trivially means "the origin stays put"; eigenvalues, however, can be 0

Figure 2 · A Set of Unit Vectors

- Unit vectors from the origin, like e_1 and e_2 , have their tips on the unit circle



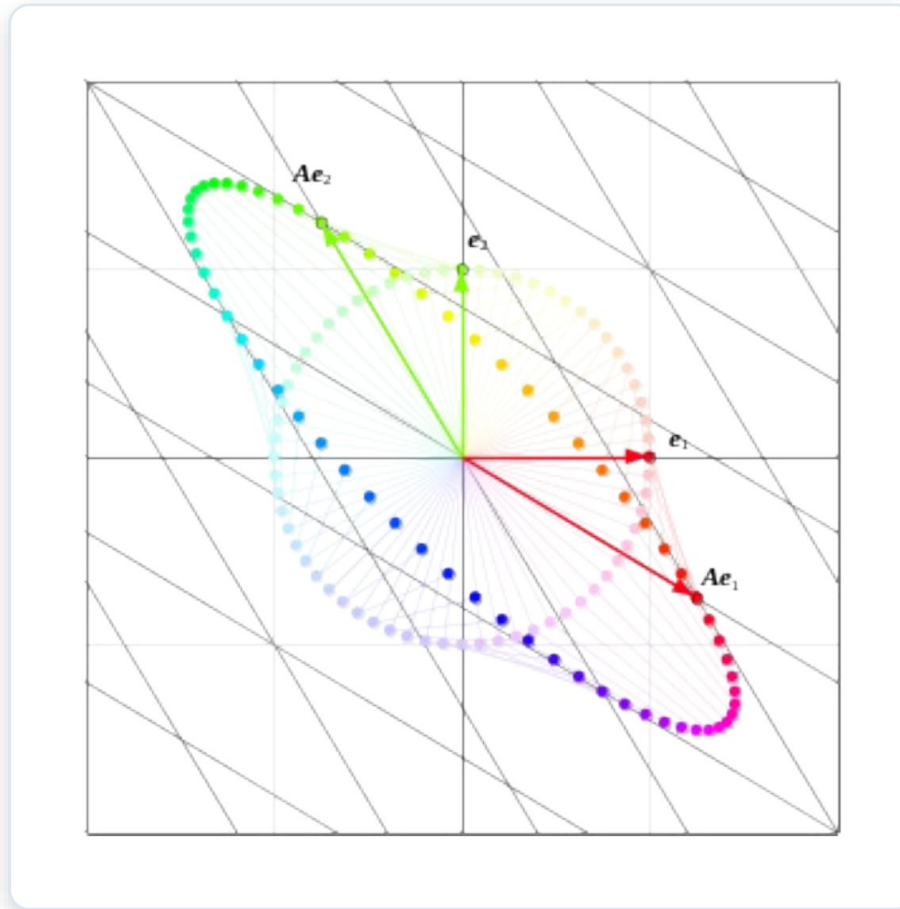
A acting on unit vectors e_1, e_2 changes both their size and direction

$$Ae_1 = [5/4; -3/4] \quad Ae_2 = [-3/4; 5/4]$$

- We track the standard basis $[e_1, e_2]$ to see the unit square grid become a parallelogram grid
- Try computing the determinant of A, and watch how the area changes

Figure 3 · Matrix A's Linear Transform of a Set of Unit Vectors

- The unit circle becomes a rotated ellipse; the unit square grid tilts into a parallelogram grid



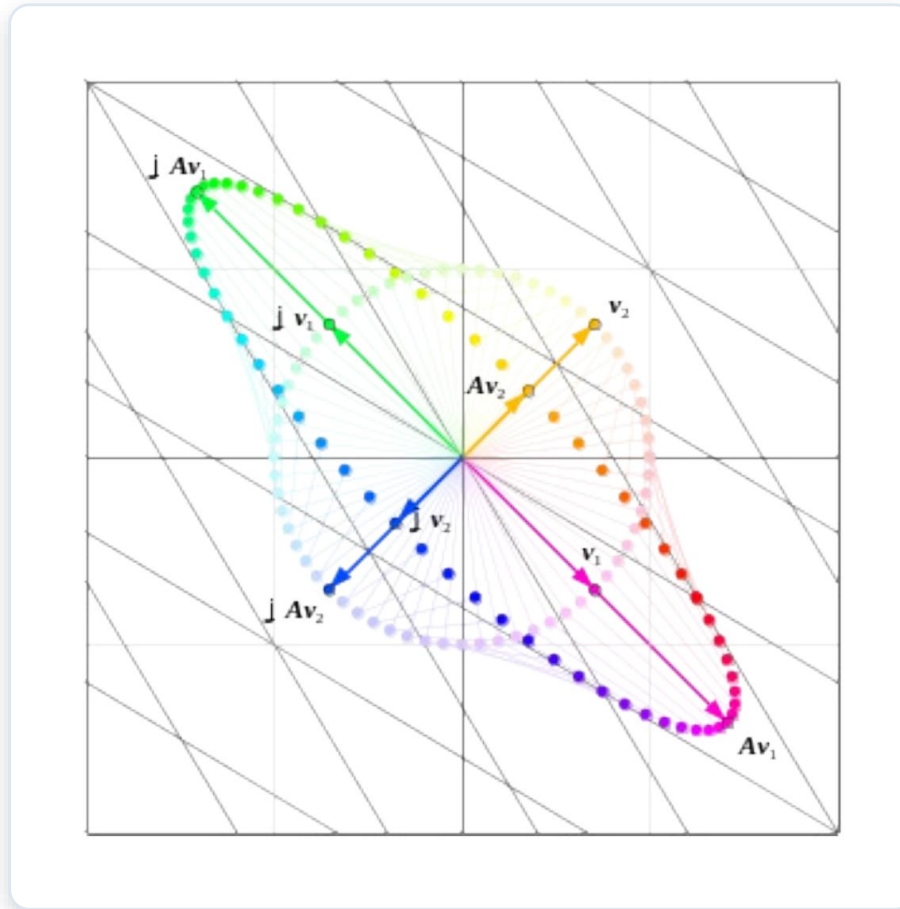
A closer look reveals special vectors whose direction stays fixed, only scaled

$$Av_1 = 2v_1$$

- Unit vector v_1 (magenta), transformed by A , keeps its direction but scales to twice its length
- This tells us v_1 is an eigenvector of A , with eigenvalue $\lambda_1=2$

Figure 4 · Matrix A's Eigenvectors

- v_1 (magenta) and v_2 (orange) keep their direction under A , with eigenvalues $\lambda_1=2$ and $\lambda_2=0.5$



EIGENVECTORS ARE NOT UNIQUE

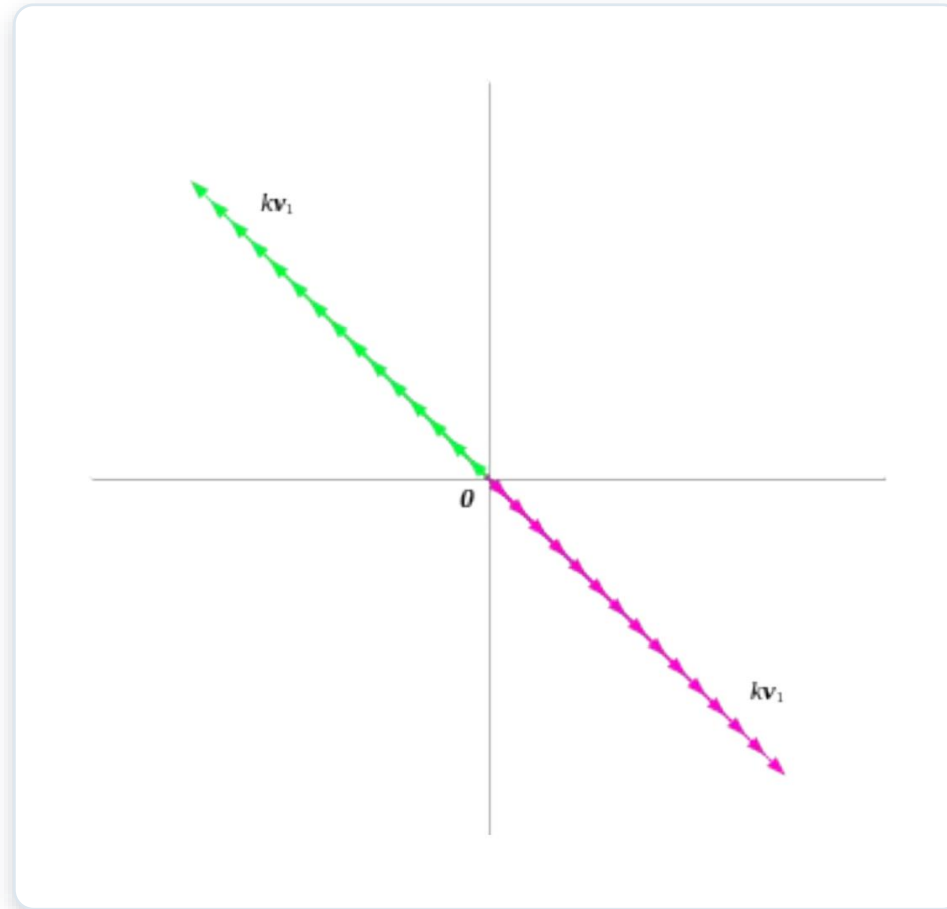
Eigenvalues and eigenvectors pair up, but an eigenvector for one λ is not unique

$$A(-\mathbf{v}_1) = 2(-\mathbf{v}_1) \quad A(k\mathbf{v}_1) = 2(k\mathbf{v}_1)$$

- The reverse $-\mathbf{v}_1$, and any nonzero scalar multiple $k \mathbf{v}_1$, are also eigenvectors for $\lambda_1=2$
- All such eigenvectors form a line through the origin; this book's eigenvectors are unit-normalized

Figure 5 · Eigenvectors Are Not Unique

- Scaling by any nonzero scalar k still gives an eigenvector for the same eigenvalue $\lambda_1=2$



ANOTHER EIGENVECTOR

$\mathbf{v}_2$ (orange) is also an eigenvector of $\mathbf{A}$, with eigenvalue $\lambda_2=0.5$

$\mathbf{A}\mathbf{v}_2=0.5\mathbf{v}_2$; try sketching the line of eigenvectors for $\lambda_2=0.5$ on Figure 5

The unit circle became a rotated ellipse! The reason behind this is covered in Chapter 13

Rearranging $Av=\lambda v$ gives $(A-\lambda I)v=0$; since v is nonzero, $A-\lambda I$ must be singular

$$(A - \lambda I)v = 0 \quad \det(A - \lambda I) = 0$$

- v is a nonzero unit vector reduced to the zero vector by $A-\lambda I$, so $A-\lambda I$ is not full rank

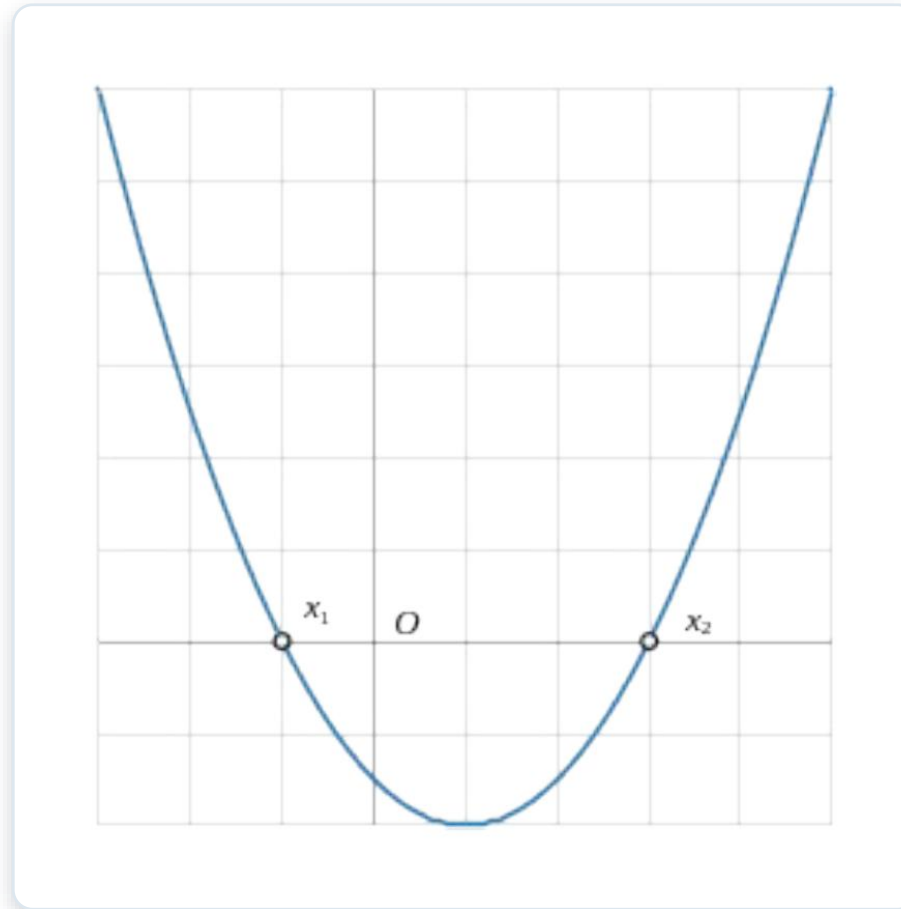
The polynomial from $\det(A-\lambda I)$ is called the characteristic polynomial

$$\lambda^2 - 5/2\lambda + 1 = 0$$

- Setting the characteristic polynomial to 0; as Figure 6 shows, a quadratic has two real roots when $b^2-4ac \geq 0$

Figure 6 · A Quadratic Function and the Roots of Its Equation

- The quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ gives the eigenvalues once substituted in



This gives A's two eigenvalues

$$\lambda_1 = 2 \quad \lambda_2 = 1/2$$

- For $\lambda_1=2$, substitute $v_1=[x_1, x_2]^T$ into $(A-\lambda I)v=0$ to get a homogeneous linear system

Simplifying gives $x_1+x_2=0$, a line through the origin

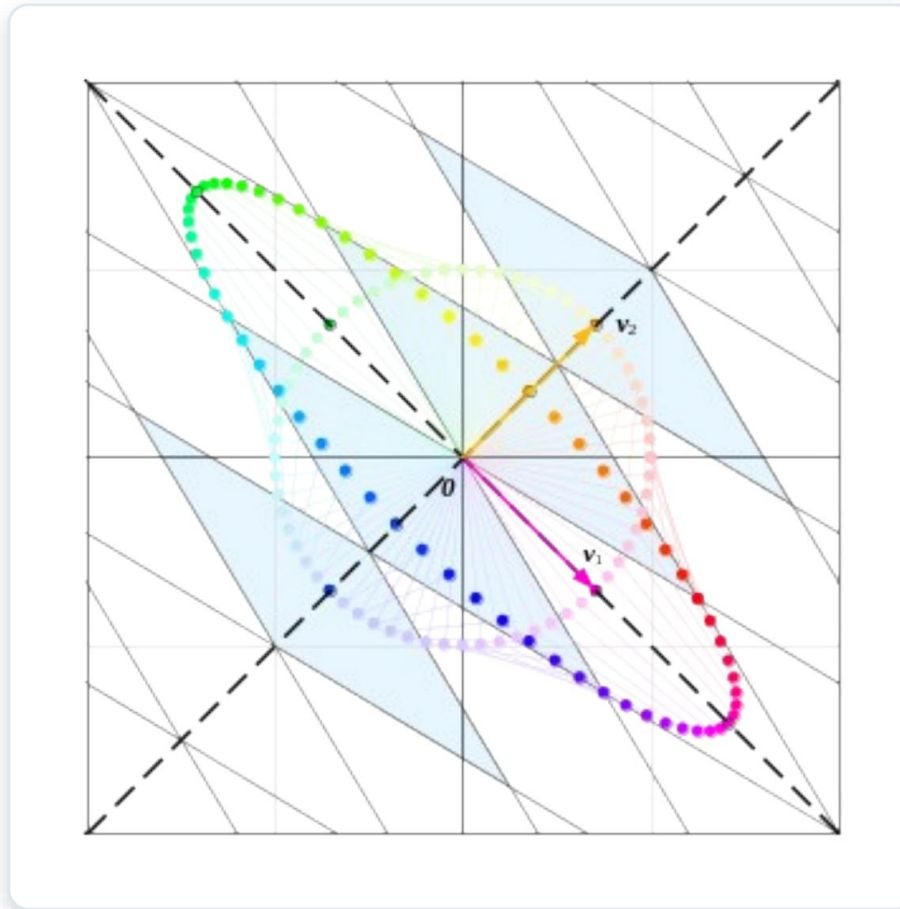
A homogeneous system has all constant terms zero; $Ax=b$ with b the zero vector always has the zero solution

Besides the zero solution, a homogeneous system can also have infinitely many nonzero solutions

Any nonzero vector on the line $x_1+x_2=0$ is an eigenvector for $\lambda_1=2$

Figure 7 · Two Lines (Through the Origin)

- $x_1+x_2=0$ and $x_1-x_2=0$ pass through the origin, right at the parallelogram grid's intersections



For $\lambda_2=0.5$, the same steps simplify to $x_1-x_2=0$

$$\mathbf{v}_1 = [\sqrt{2}/2; -\sqrt{2}/2] \quad \mathbf{v}_2 = [\sqrt{2}/2; \sqrt{2}/2]$$

- Pick $\mathbf{v}_1$ in the fourth quadrant and $\mathbf{v}_2$ in the first, so $\mathbf{v}_1$ leads $\mathbf{v}_2$ counterclockwise and $\det[\mathbf{v}_1, \mathbf{v}_2] > 0$

Combining both eigen-equations gives $AV=V\Lambda$; if V is full rank, A can be written

$$A = V\Lambda V^{-1}$$

- Right-multiplying V by the diagonal matrix Λ scales V 's columns
- A matrix that admits this form is called a diagonalizable matrix

Having eigenvalues and eigenvectors does not guarantee diagonalizability

Linearly independent eigenvectors are the key condition — that is, V must be invertible

A 's columns being linearly independent does not imply A 's eigenvectors are linearly independent

Only when the eigenvectors are independent does V have an inverse, giving $A = V\Lambda V^{-1}$

ROTATE → SCALE → ROTATE

Here V is exactly the rotation matrix covered earlier, and V is orthogonal

$$V^{-1} = V^T \quad A = V\Lambda V^T$$

- This example's eigendecomposition is geometrically "rotate → scale → rotate"
- This special structure arises exactly because A is a symmetric matrix

A symmetric matrix's eigendecomposition has its own name — spectral decomposition

Spectral decomposition refers specifically to the eigendecomposition of a real symmetric matrix — next section's topic

Note: an asymmetric matrix's eigendecomposition is not "rotate → scale → rotate"

Computing Eigenvalue Decomposition in Python

`lambdas, V = np.linalg.eig(A)`
computes eigenvalues and eigenvectors

- (a) `numpy.linalg.eig` computes A's eigenvalues/vectors; `numpy.diag` builds Λ
- (b) Take the first eigenvalue from the list
- (c) Take column 0 of V as the first eigenvector; `[:, [0]]` keeps it a 2D column
- (d)(e) Repeat the same steps for the second eigenvalue and eigenvector
- (f) Check with `V@Λ@inv(V)`; here V is invertible, so A is diagonalizable

```
## 初始化
import numpy as np
import matplotlib.pyplot as plt

## 构造矩阵
A = np.array([[1.25, -0.75],
              [-0.75, 1.25]])

## 特征值分解
a lambdas, V = np.linalg.eig(A)
  Lambda = np.diag(lambdas)

## 第一组特征值、特征向量
# 特征值
b lambda_1 = lambdas[0]
# 特征向量
c v_1 = V[:, [0]]
  A @ v_1
  lambda_1 * v_1

## 第二组特征值、特征向量
# 特征值
d lambda_2 = lambdas[1]

# 特征向量
e v_2 = V[:, [1]]
  A @ v_2
  lambda_2 * v_2

## 特征值分解
f V @ Lambda @ np.linalg.inv(V)
```

SUMMARY

What Have We Seen About Eigenvalue Decomposition?

Eigendecomposition finds A 's eigenvalue λ and eigenvector v satisfying $Av = \lambda v$

An eigenvector keeps its direction under A ; the eigenvalue sets its scale and sign

Solve for eigenvalues by setting the characteristic polynomial $\det(A - \lambda I) = 0$

Solve for eigenvectors by substituting λ into the homogeneous system $(A - \lambda I)v = 0$

When eigenvectors are independent, A is diagonalizable: $A = V\Lambda V^{-1}$

A symmetric matrix's eigendecomposition is "rotate $\rightarrow$ scale $\rightarrow$ rotate" — spectral decomposition

Key Terms in This Section

Eigenvalue Decomposition *EVD — finds a matrix's eigenvalues and eigenvectors*

Eigenvector *v — a nonzero vector whose direction is unchanged, taken as a unit vector*

Homogeneous System *A linear system $Ax=0$ with all constant terms zero*

Diagonal Matrix Λ *The diagonal matrix formed from the eigenvalues*

Rotation / Orthogonal Matrix *The geometric identity of V in spectral decomposition, $V^{-1}=V^T$*

Eigenvalue *λ — the factor by which an eigenvector is scaled; can be 0*

Characteristic Polynomial *The polynomial from $\det(A-\lambda I)=0$*

Diagonalizable Matrix *A matrix that can be written as $A=V\Lambda V^{-1}$*

Spectral Decomposition *The eigendecomposition of a real symmetric matrix, $A=V\Lambda V^T$*

Determinant *The factor by which a linear transform scales area or volume*

Further Reading

Further Reading · 11.2 Spectral Decomposition

The next section systematically covers real symmetric matrices' eigendecomposition

Further Reading · 11.3 Asymmetric Matrix EVD

An asymmetric matrix's decomposition is not "rotate $\rightarrow$ scale $\rightarrow$ rotate"

Review · Rotation and Orthogonal Matrices

The geometric intuition here builds on orthogonal matrices — worth revisiting

Open-Source Repository

Code and figures for the whole book: <https://github.com/Visualize-ML>

Exercises Q1 – Q9

Q1a By hand, find the eigendecomposition of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, and verify with Python.

Q1c By hand, find the eigendecomposition of $\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$, and verify with Python.

Q1e By hand, find the eigendecomposition of $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, and verify with Python.

Q2 State which matrices in Q1 are diagonalizable and which are not, with reasons.

Q4 Prove that all eigenvectors for one eigenvalue, together with the zero vector, form a subspace.

Q1b By hand, find the eigendecomposition of $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, and verify with Python.

Q1d By hand, find the eigendecomposition of $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$, and verify with Python.

Q1f By hand, find the eigendecomposition of $\begin{bmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$, and verify with Python.

Q3 Why is an eigenvector generally required to be nonzero? Must an eigenvalue also be nonzero? Give an example.

Exercises Q10 – Q17

Q10 Write the general characteristic polynomial of a 2×2 matrix A in terms of its trace and determinant.

Q12 Build a 2×2 symmetric matrix, solve its eigendecomposition by hand, and check the eigenvectors are orthogonal.

Q14 Build a 2×2 matrix with a repeated eigenvalue, and check whether it is still diagonalizable.

Q16 Prove: if v is an eigenvector of A , then v is also an eigenvector of A^2 , with eigenvalue λ^2 .

Q11 Prove: the sum of A 's eigenvalues equals its trace, and their product equals its determinant.

Q13 Write a Python function `eig_2x2(A)` that solves a 2×2 matrix's eigenvalues by hand via the characteristic polynomial.

Q15 Use matplotlib to plot A 's transform of the unit circle, marking the eigenvector directions.

Q17 Explore: does a rotation matrix (with no real eigenvalues) have real eigenvectors at all?

Exercises Q18 – Q25

Q18 Use `numpy.linalg.eig` to check whether your hand solution to Q12 agrees.

Q20 Write a Python function `is_diagonalizable(A)` that checks if the eigenvector matrix is invertible.

Q22 Build a singular (rank-deficient) matrix, and check whether 0 is always among its eigenvalues.

Q24 Explore: if A is invertible, how do the eigenvalues of A^{-1} relate to those of A ? Prove and verify in code.

Q19 Explore: what pattern do triangular matrices' eigenvalues follow? Verify in code.

Q21 Explore: what are the identity matrix I 's eigenvalues and eigenvectors? What is special about them?

Q23 Practice: write a Python function `verify_evd(A, lambdas, V)` checking $A@V$ equals $V@diag(lambdas)$.

Q25 Preview: in the next section on spectral decomposition, why are a symmetric matrix's eigenvectors naturally orthogonal?

10 Prompts to Explore with AI

1 Explain the geometric meaning of eigenvalues and eigenvectors, and why an eigenvector is a matrix's "own direction".

2 Explain why an eigenvector must be nonzero, while an eigenvalue can be 0.

3 Explain how the characteristic polynomial $\det(A-\lambda I)=0$ is derived, step by step, with its geometric meaning.

4 Use Python to show how to hand-code solving for a 2×2 matrix's eigenvalues and eigenvectors.

5 Explain why eigenvectors for one eigenvalue are not unique, and what geometric shape they form.

10 Prompts to Explore with AI

6 Explain the condition for a matrix to be diagonalizable, and why independent eigenvectors are the key.

7 Explain why a symmetric matrix's eigendecomposition can be read as "rotate $\rightarrow$ scale $\rightarrow$ rotate".

8 Explain the difference between spectral decomposition and general eigendecomposition, and why it needs symmetry.

9 Explain the relationship between $A=V\Lambda V^{-1}$ and the matrix equation $AV=V\Lambda$.

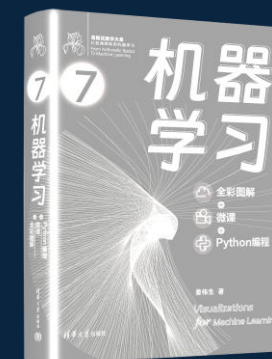
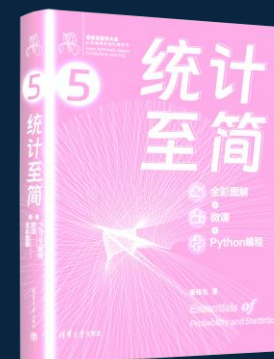
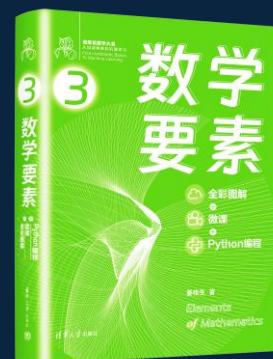
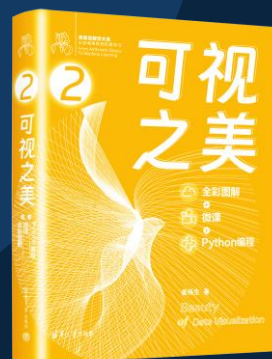
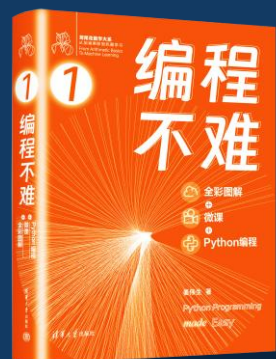
10 Use Python to visualize a 2×2 matrix's transform of the unit circle, marking its eigenvector directions.

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

